Department of Mechanical and Production Engineering
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Homework Exercises

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Lecture 1: Introduction

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Problem 1.1a

Show that

$$\delta_{ij}\delta_{ij} = 3.$$

Derivation

Expanding $\delta_{ij}\delta_{ij}$ we get

$$\delta_{ij}\delta_{ij} = \delta_{11}\delta_{11} + \delta_{12}\delta_{12} + \delta_{13}\delta_{13} + \delta_{21}\delta_{21} + \delta_{22}\delta_{22} + \delta_{23}\delta_{23} + \delta_{31}\delta_{31} + \delta_{32}\delta_{32} + \delta_{33}\delta_{33}.$$

As $\delta_{11} = \delta_{22} = \delta_{33} = 1$ the above becomes:

$$\delta_{ij}\delta_{ij} = 1 \cdot 1 + 1 \cdot 1 + 1 \cdot 1 = 3.$$

$$\boxed{\delta_{ij}\delta_{ij} = 3}$$ RESULT

Problem 1.1c

Show that

$$\alpha_{ij} b_{jk} = \alpha_{in} b_{nk}.$$

Derivation

We expand by summing over the common index j:

$$\alpha_{ij} b_{jk} = \sum_j \alpha_{ij} b_{jk} = \alpha_{i1} b_{1k} + \alpha_{21} b_{2k} + \alpha_{i3} b_{3k}.$$

Now we sum over n instead:

$$\alpha_{i1} b_{1k} + \alpha_{i2} b_{2k} + \alpha_{i3} b_{3k} = \sum_n \alpha_{in} b_{nk}.$$

Q.E.D.

$$\boxed{\alpha_{ij} b_{jk} = \alpha_{in} b_{nk}}$$

RESULT

Problem 1.1e

Show that

$$\delta_{ij} C_{jkl} = C_{ikl}.$$

Derivation

Expanding we get

$$\delta_{ij} C_{jkl} = \delta_{11} C_{1kl} + \delta_{22} C_{2kl} + \delta_{33} C_{3kl} = C_{ikl}.$$

$$\boxed{\delta_{ij} C_{jkl} = C_{ikl}}$$

RESULT